Module 3.2

Basic concepts of phylogenetics

INFORMATION

Lecturer

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Co-lecturer(s)

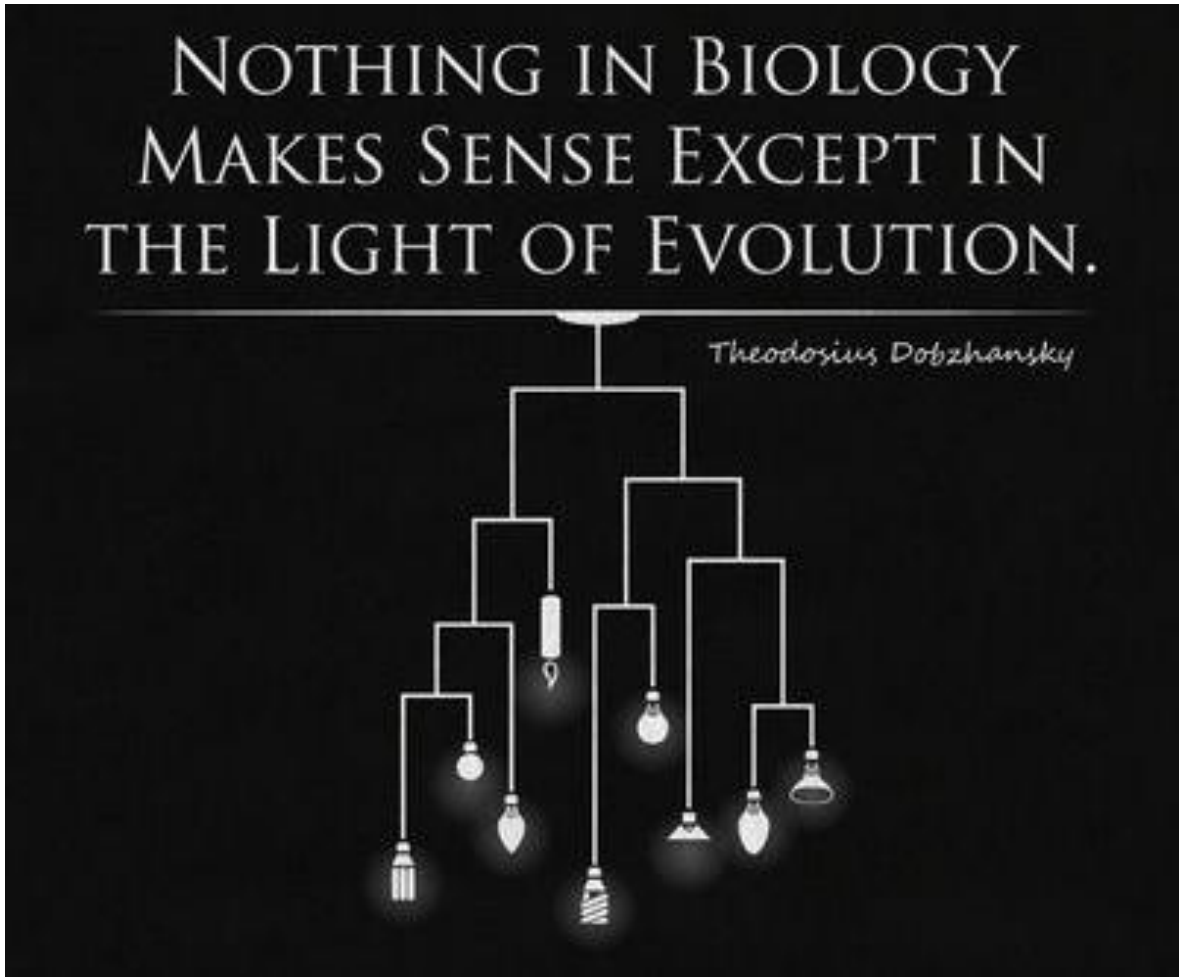
Teaching assistant

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Agenda

1. Trees: definition and importance
2. Theory – Genetic data and tree
3. Theory – Tree building, bootstrapping
4. Exercise – Tree interpretation
5. Practical – Build trees for ST147
6. Visualizing trees with metadata
7. Using tree as evidence

Why evolution?

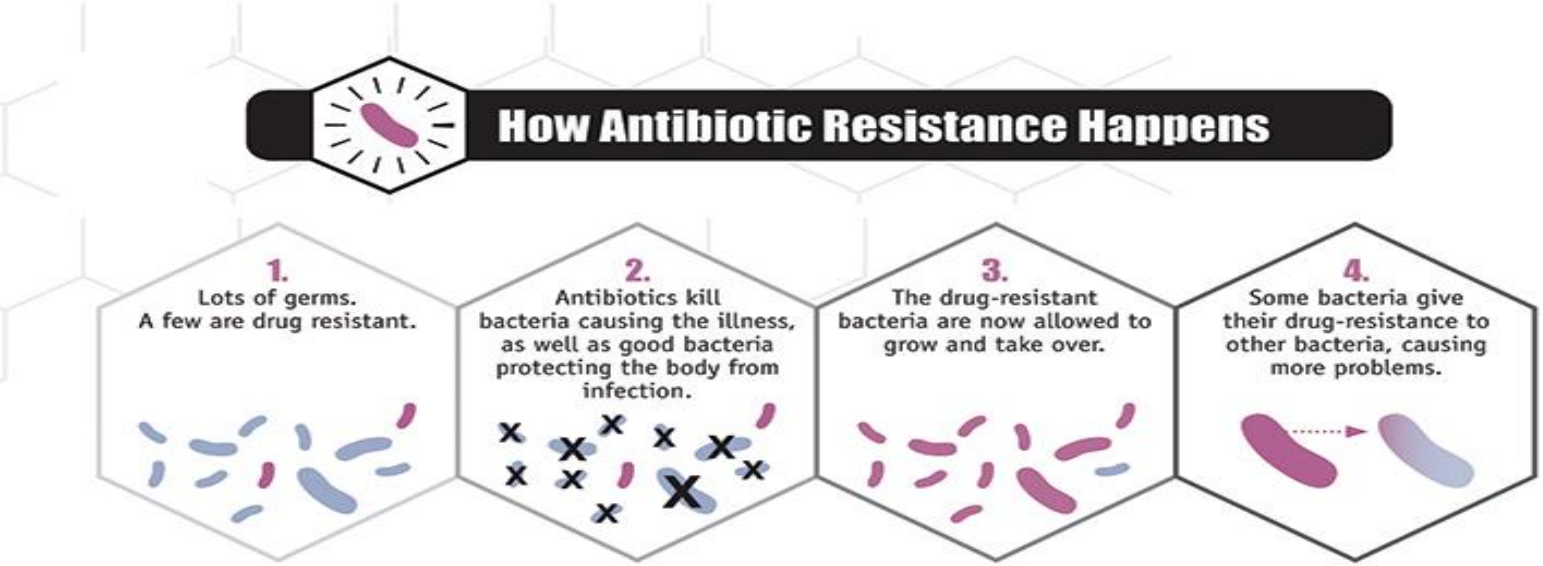


Universality

Diversity

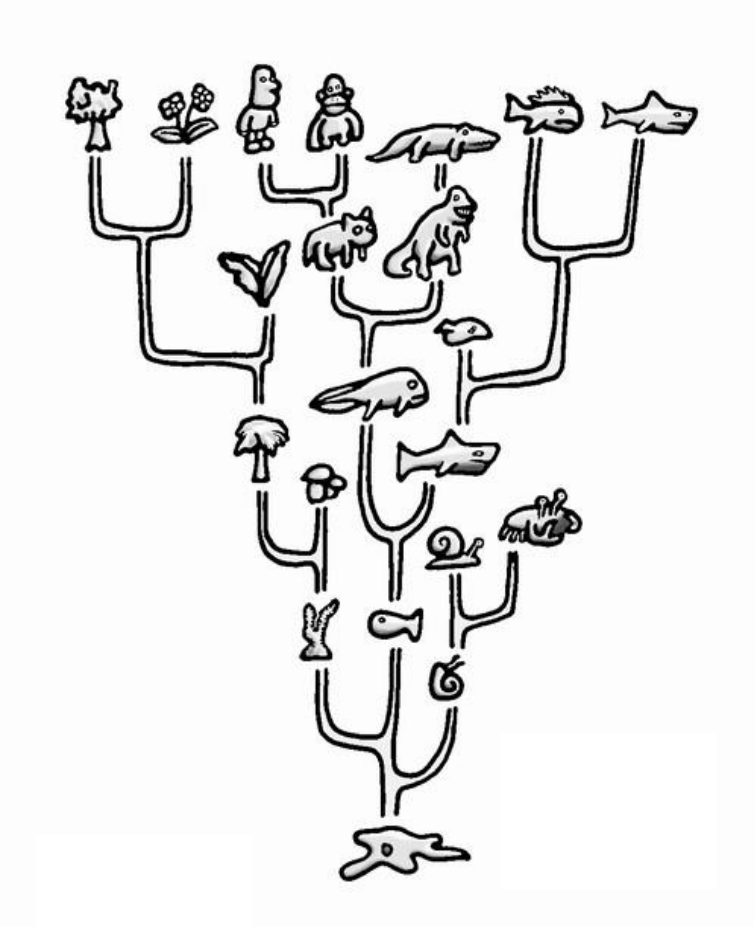
The processes of evolution

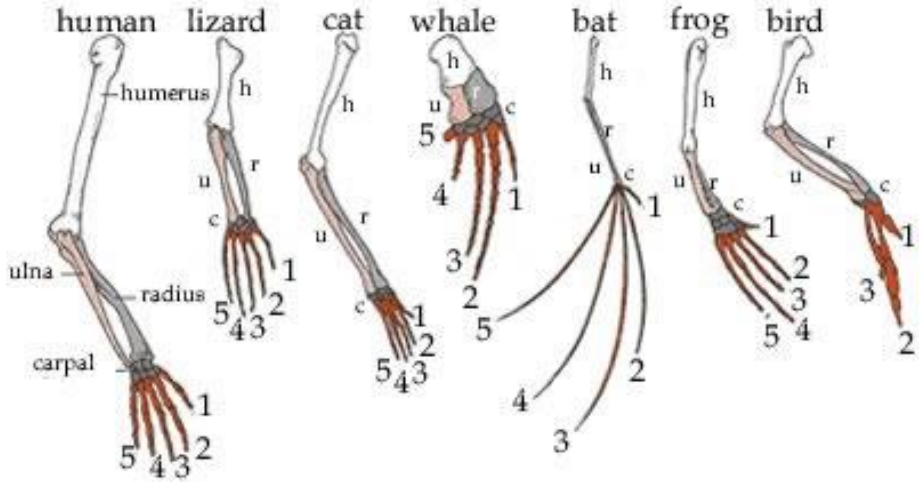
- **Mutation**: Genetic variability that evolution will act upon.
- **Natural selection**: Differential survival and reproduction of individuals due to differences in phenotype. Under natural selection, mutations could be beneficial, deleterious, or neutral.



What is a phylogeny?

- A phylogeny is a diagram describing the ancestral relationships of organisms.
- Phylogenetics looks for homology as evidence of common ancestry





Anatomical homology of forelimb in animals.

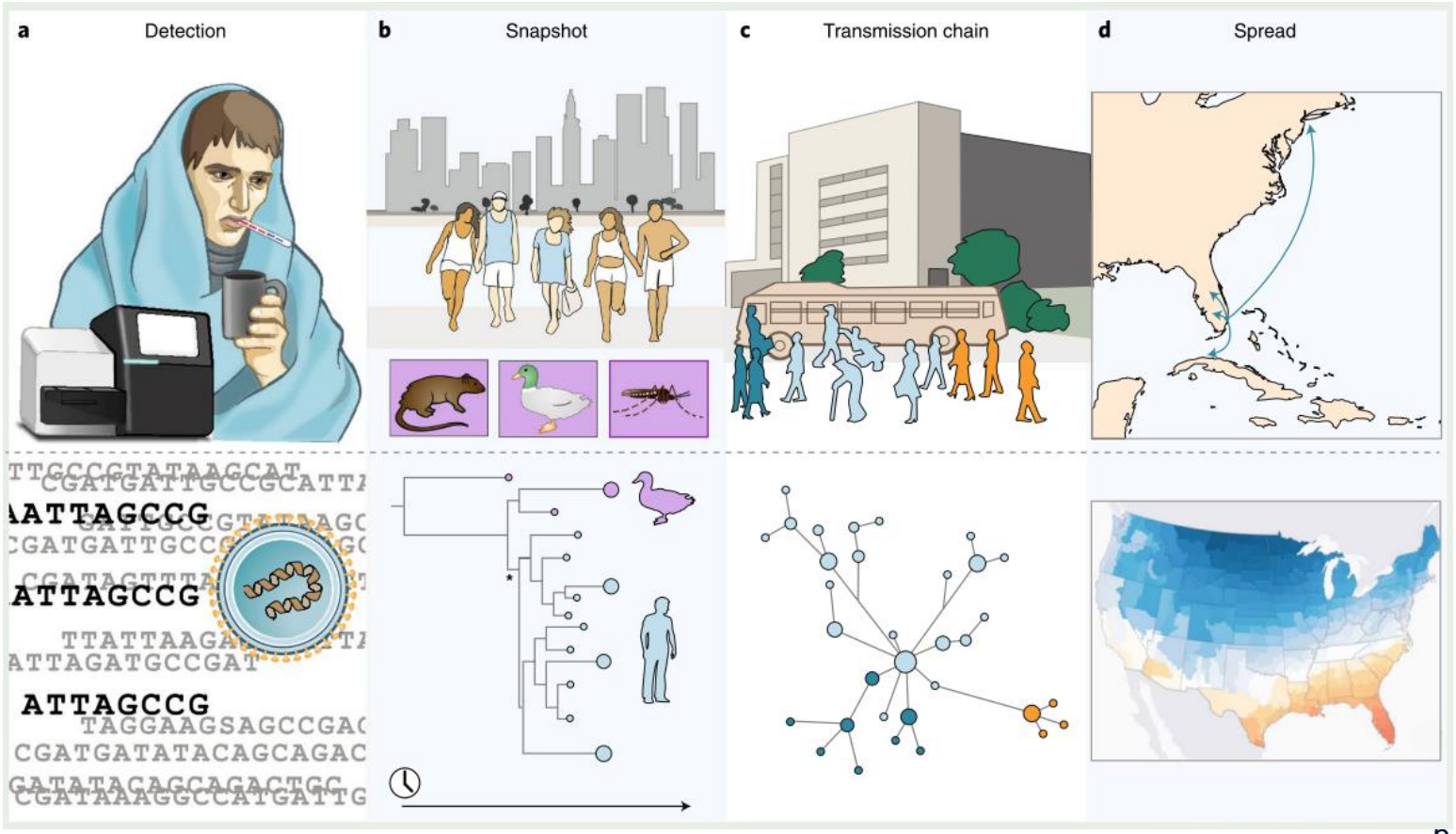
Homology: similarity among organisms due to inheritance from a shared common ancestor.

Why use phylogenetics?

- The phylogeny is (nearly) the most complete representation of evolutionary relationship.
- Provide an important evolutionary framework to understand biology, to ask questions.

For epidemiology:

- Confidently differentiate organisms at strain-level
- Provide information on transmission, clustering, epidemiology.



Homology in genetic data

- ***Similarity in nucleotide sequence almost always points to homology.***
- Constructing tree using microbe's **homology** sequence data to infer evolutionary relationships

```
SIVcpz    ATGGGTGCGA GAGCGTCAGT TCTAACAGGG GGAAAATTAG ATCGCTGGGA
HIV-1     ATGGGTGCGA GAGCGTCAGT ATTAAGCGGG GGAGAATTAG ATCGATGGGA

SIVcpz    AAAAGTTCGG CTTAGGCCCG GGGGAAGAAA AAGATATATG ATGAAACATT
HIV-1     AAAAATTCGG TTAAGGCCAG GGGGAAAGAA AAAATATAAA TTAAAACATA

SIVcpz    TAGTATGGGC AAGCAGGGAG CTGGAAAGAT TCGCATGTGA CCCCGGGCTA
HIV-1     TAGTATGGGC AAGCAGGGAG CTAGAACGAT TCGCAGTTAA TCCTGGCCTG

SIVcpz    ATGGAAAGTA AGGAAGGATG TACTAAATTG TTACAACAAT TAGAGCCAGC
HIV-1     TTAGAAACAT CAGAAGGCTG TAGACAAATA CTGGGACAGC TACAACCATC

SIVcpz    TCTCAAAACA GGCTCAGAAG GACTGCGGTC CTTGTTTAAC ACTCTGGCAG
HIV-1     CCTTCAGACA GGATCAGAAG AACTTAGATC ATTATATAAT ACAGTAGCAA

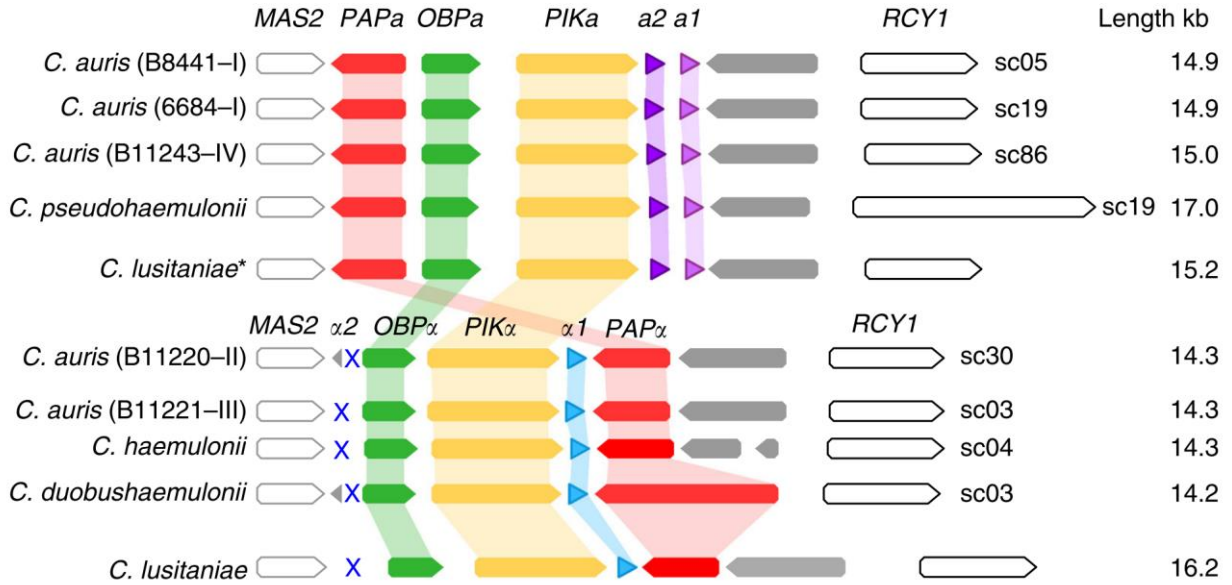
SIVcpz    TACTGTGGTG CATACATAGT GACATCACTG TAGAAGACAC ACAGAAAGCT
HIV-1     CCCTCTATTG TGTGCATCAA AGGATAGAGA TAAAAGACAC CAAGGAAGCT

SIVcpz    CTAGAACAGC TAAAGCGGCA TCATGGAGAA CAACAGAGCA AAACTGAAAG
HIV-1     TTAGACAAGA TAGAG--GAA -----GAGCA AAACAAAAGT AA---GAAAA

SIVcpz    TAACTCAGGA AGCCGTGAAG GGGGAGCCAG TCAAGGCGCT AGTGCCTCTG
HIV-1     AAGCACAGCA AGC-----AG CAGCTGACA- -CAGGACAC- AG--CAGC--

SIVcpz    CTGGCATTAG TGGAAATTAC
HIV-1     CAGG--TCAG CCAAAATTAC
```

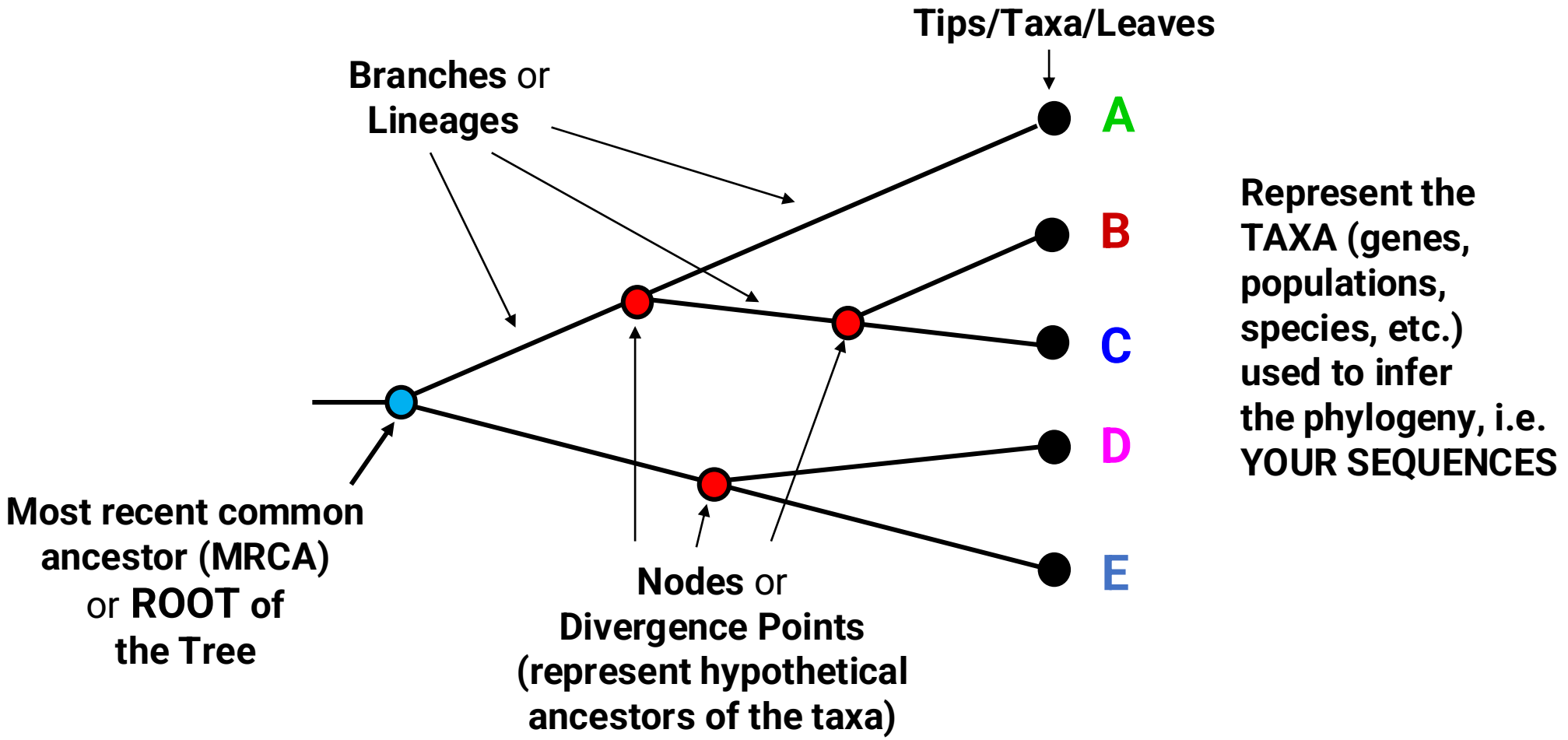
Genetic comparison between HIV-1 and Simian Immunodeficiency Virus (SIV)



Genomic organization in genomes of multiple *Candida* species


How to read a tree?

Some terms

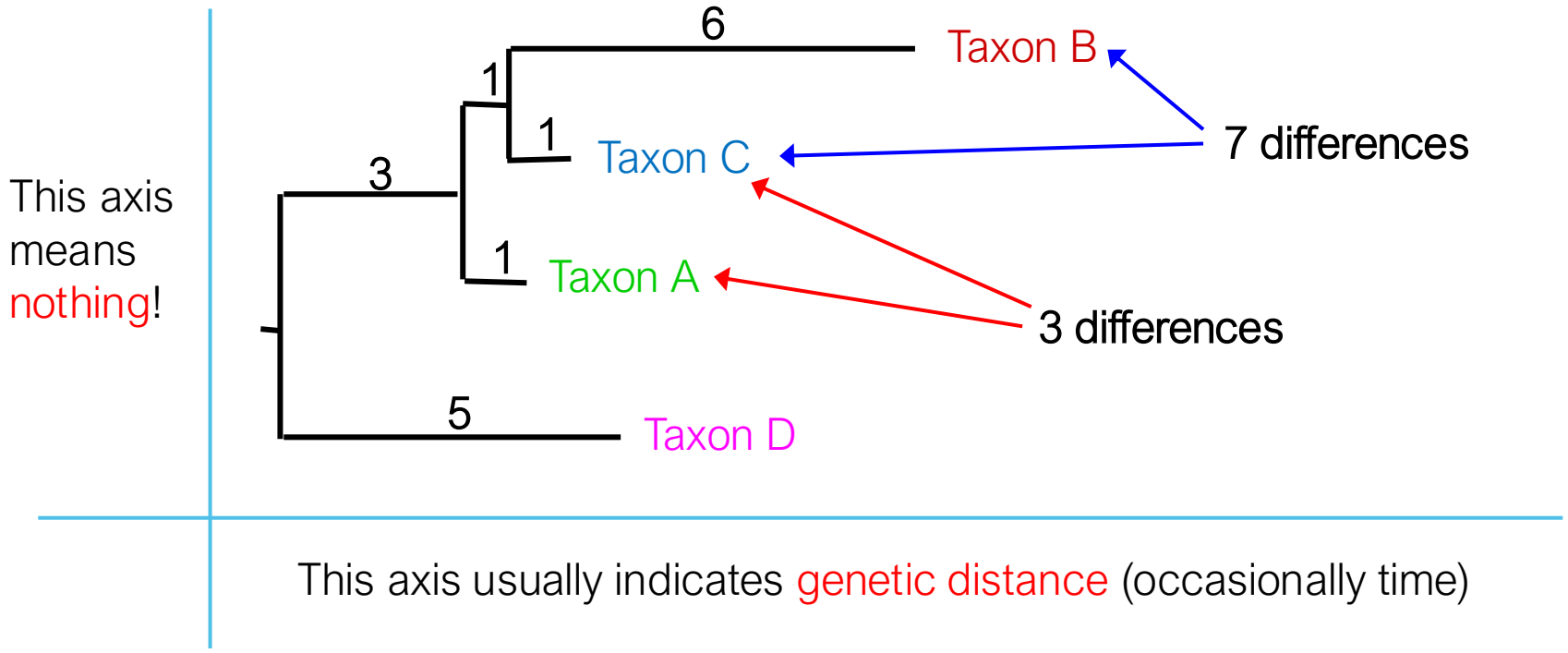


Represent the TAXA (genes, populations, species, etc.) used to infer the phylogeny, i.e. YOUR SEQUENCES

Similarity vs. relatedness

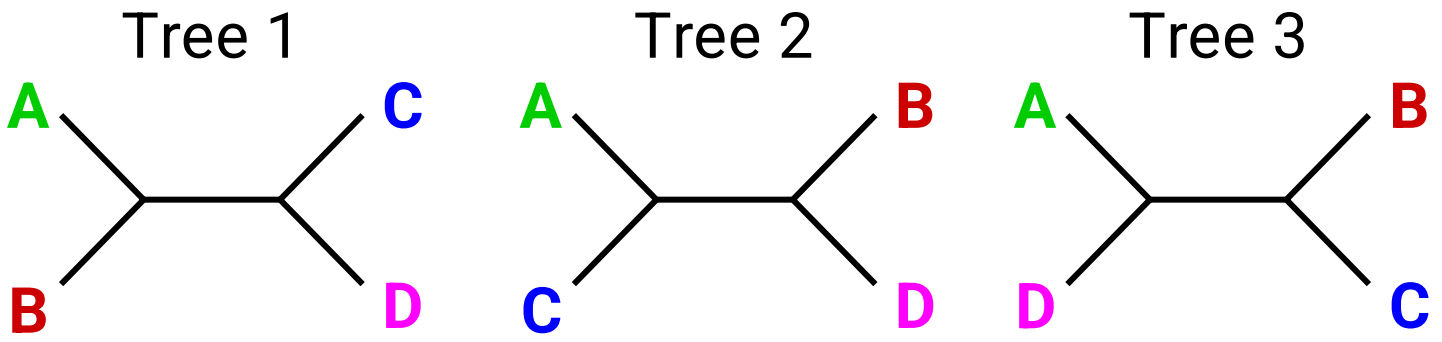
Sequence similarity and relatedness are not the same thing, even though evolutionary relationships are based on certain types of similarity

- Similarity: being similar in a measurable metric (substitution differences)
- Relatedness: genetically/evolutionarily connectedness (historical fact)



Unrooted trees

The possible unrooted trees of four taxa (A, B, C, D)



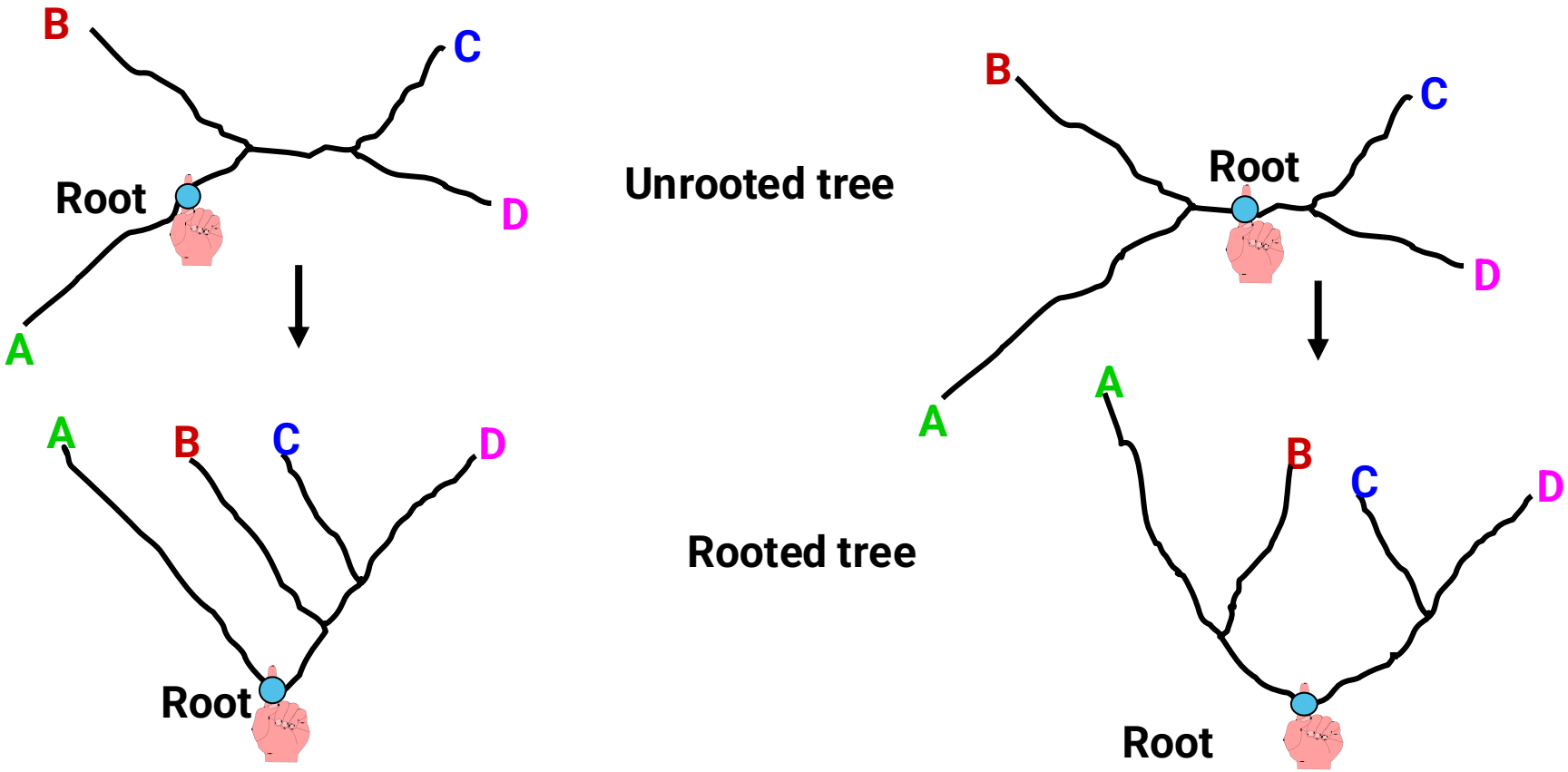
- Unrooted trees tell us the similarity among taxa, but not the ancestry nor the origin

- The number of unrooted trees increases in a greater than exponential manner with the number of taxa

# Taxa (N)	# Unrooted trees
3	1
4	3
5	15
6	105
7	945
8	10,935
9	135,135
10	2,027,025
.	.
.	.
.	.
.	.
30	$\approx 3.58 \times 10^{36}$

Finding a root

Inferring evolutionary relationship requires a rooted phylogeny

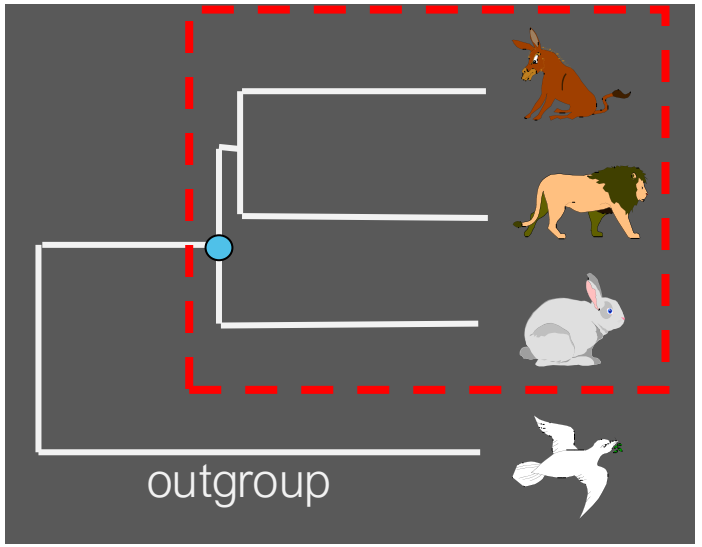


- Both tree are *technically* correct, but give us **different** evolutionary stories
- Finding a *correct* root is important and also difficult.

Finding a root

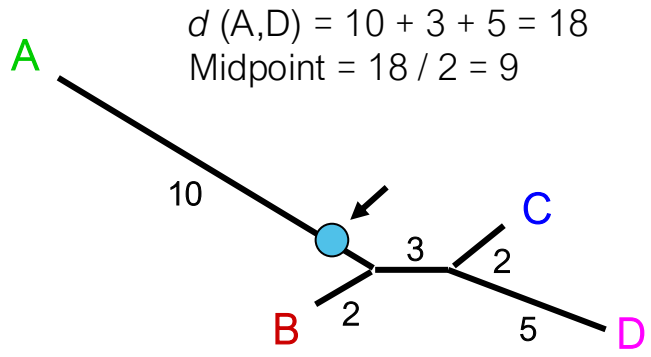
By outgroup:

- Use taxa (outgroup) that are known to fall outside of the group of interest
- Require some prior knowledge about the relationship between taxa



Monophyletic clade

Outgroup/Basal taxon



By midpoint or distance:

- Default by many software
- Root the tree at the midway point between the most two distant taxa.
- OFTEN UNRELIABLE

From sequences to tree

1. Formulate a hypothesis!

2. Gather **appropriate** sequences
   - From your samples
   - From publicly available databases

3. Align sequences
   - Do you have an appropriate outgroup?

4. Run preliminary trees

5. Then run more trees, test, and run trees, and further analyses

DNA sequence alignment

- DNA sequence alignment (input) is the most important thing in phylogenetic reconstruction.
- Arrange sequences to identify regions of similarity that may be a consequence of homologous relationship.

```
Taxon 1   GCGGCCCA TCAGGTAGTT GGTGG
Taxon 2   GCGGCCCA TCAGGTAGTT GGTGG
Taxon 3   GCGTTCCA TCAGCTGGTT GGTGG
Taxon 4   GCGTCCCA TCAGCTAGTT GGTGG
Taxon 5   GCGGCGCA TTAGCTAGTT GGTGA
          ******** **********
          *****
          TTGACATG CCGGGG---A AACCG
          TTGACATG CCGGTG--GT AAGCC
          TTGACATG -CTAGG---A ACGCG
          TTGACATG -CTAGGGAAC ACGCG
          TTGACATC -CTCTG---A ACGCG
          ******** ??????????
          *****
```

Easy

Difficult due to insertions or deletions (indels)

Estimating genetic distances

Simplest way: Genetic distance is the **observed mutational differences** between taxa

```
SIVcpz   ATGGGTGCGA GAGCGTCAGT TCTAACAGGG GGAAAATTAG ATCGCTGGGA
HIV-1    ATGGGTGCGA GAGCGTCAGT ATTAAGCGGG GGAGAATTAG ATCGATGGGA

SIVcpz   AAAAGTTCGG CTTAGGCCCG GGGGAAGAAA AAGATATATG ATGAAACATT
HIV-1    AAAAATTCGG TTAAGGCCAG GGGGAAAGAA AAAATATAAA TTAAAACATA

SIVcpz   TAGTATGGGC AAGCAGGGAG CTGGAAAGAT TCGCATGTGA CCCCGGGCTA
HIV-1    TAGTATGGGC AAGCAGGGAG CTAGAACGAT TCGCAGTTAA TCCTGGCCTG

SIVcpz   ATGGAAAGTA AGGAAGGATG TACTAAATTG TTACAACAAT TAGAGCCAGC
HIV-1    TTAGAAACAT CAGAAGGCTG TAGACAAATA CTGGGACAGC TACAACCATC

SIVcpz   TCTCAAAACA GGCTCAGAAG GACTGCGGTC CTTGTTTAAC ACTCTGGCAG
HIV-1    CCTTCAGACA GGATCAGAAG AACTTAGATC ATTATATAAT ACAGTAGCAA

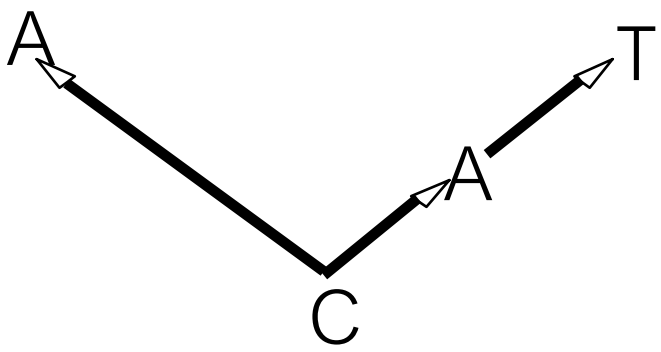
SIVcpz   TACTGTGGTG CATACATAGT GACATCACTG TAGAAGACAC ACAGAAAGCT
HIV-1    CCCTCTATTG TGTGCATCAA AGGATAGAGA TAAAAGACAC CAAGGAAGCT

SIVcpz   CTAGAACAGC TAAAGCGGCA TCATGGAGAA CAACAGAGCA AAACTGAAAG
HIV-1    TTAGACAAGA TAGAG--GAA -----GAGCA AAACAAAAGT AA---GAAAA

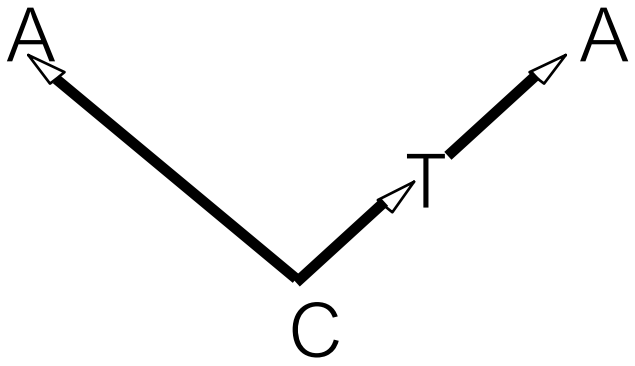
SIVcpz   TAACTCAGGA AGCCGTGAAG GGGGAGCCAG TCAAGGCGCT AGTGCCTCTG
HIV-1    AAGCACAGCA AGC-----AG CAGCTGACA- -CAGGACAC- AG--CAGC--

SIVcpz   CTGGCATTAG TGGAAATTAC
HIV-1    CAGG--TCAG CCAAAATTAC
```

DNA sequence alignment between HIV-1 and SIV.



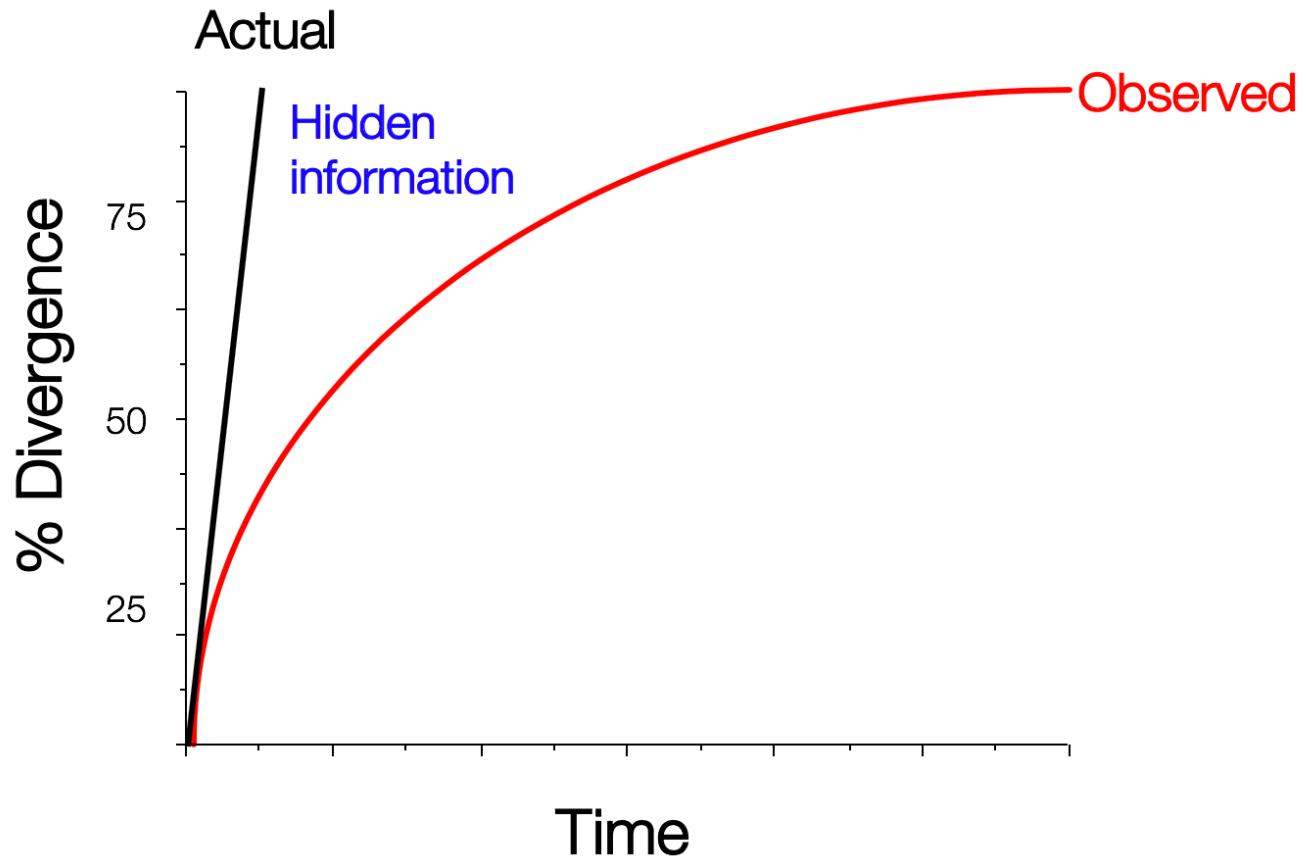
Count 1 mutations when 3 have occurred



Count 0 mutation when 3 have occurred.

Multiple substitutions at a single site – hidden information.

The problem of multiple substitutions



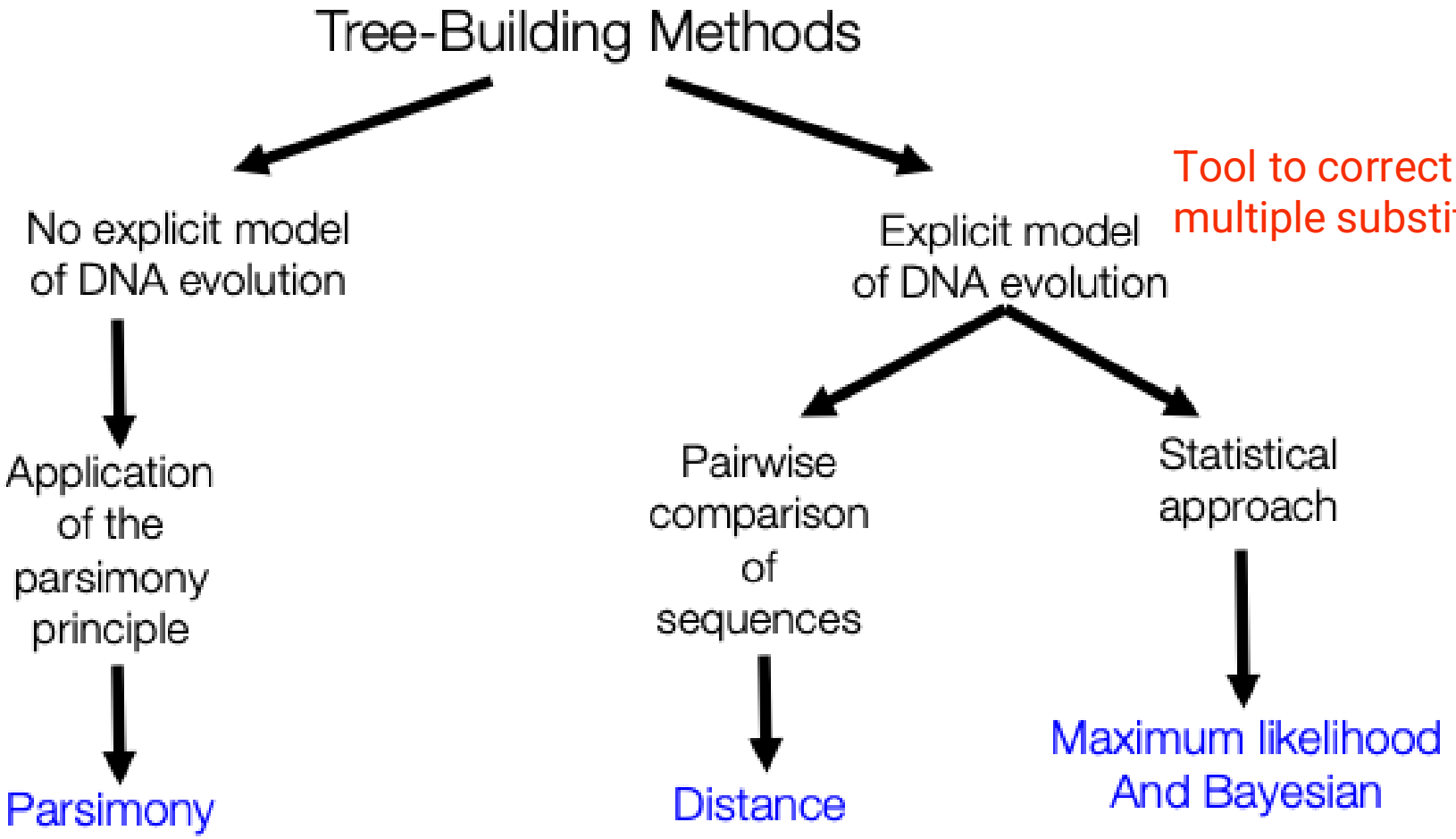
- When % divergence is low, observed distance (*p*) is a good estimator of genetic distance (*d*)
- When % divergence is high, *p* underestimates *d* and a correction statistic is required

Brief summary

- Phylogenetic trees are (nearly) best model to represent genetic relatedness and evolutionary history of a groups of organisms
- Trees are built using **homologous** sequence alignments
- Unrooted trees provide information about relatedness, while rooted trees tell more complete story of evolution



How to build a tree?


Tree-Building Methods
No explicit model of DNA evolution
Application of the parsimony principle
Parsimony
Explicit model of DNA evolution
Pairwise comparison of sequences
Distance
Statistical approach
Maximum likelihood And Bayesian
Tool to correct multiple substitutions
Most often seen in publications

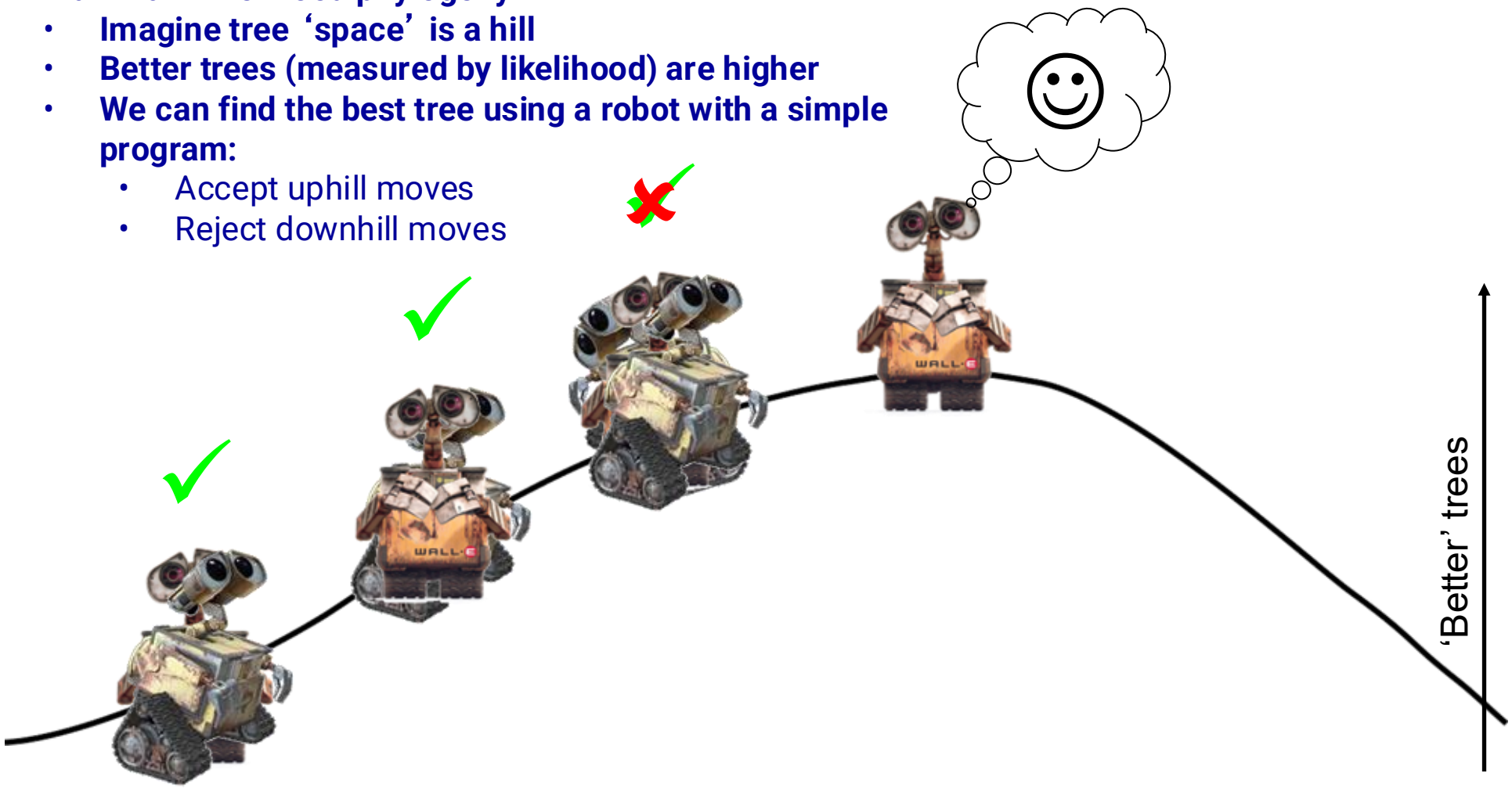
Method	Data used	Tree search	Evolutionary Model	Case use
Distance	Pairwise distance	Simple algorithm	Can be complex	For preliminary tree
Parsimony	All sites in alignment	Hill climbing	Simple	When the expected divergence is minimal
Maximum likelihood	**All sites**	**Hill climbing**	**Can be complex**	**Default option, suitable to most situations**
Bayesian Methods	All sites (+ other info)	MCMC	Can be very complex	When extra analyses are required

Hill Climbing

Statistical phylogenetic reconstruction

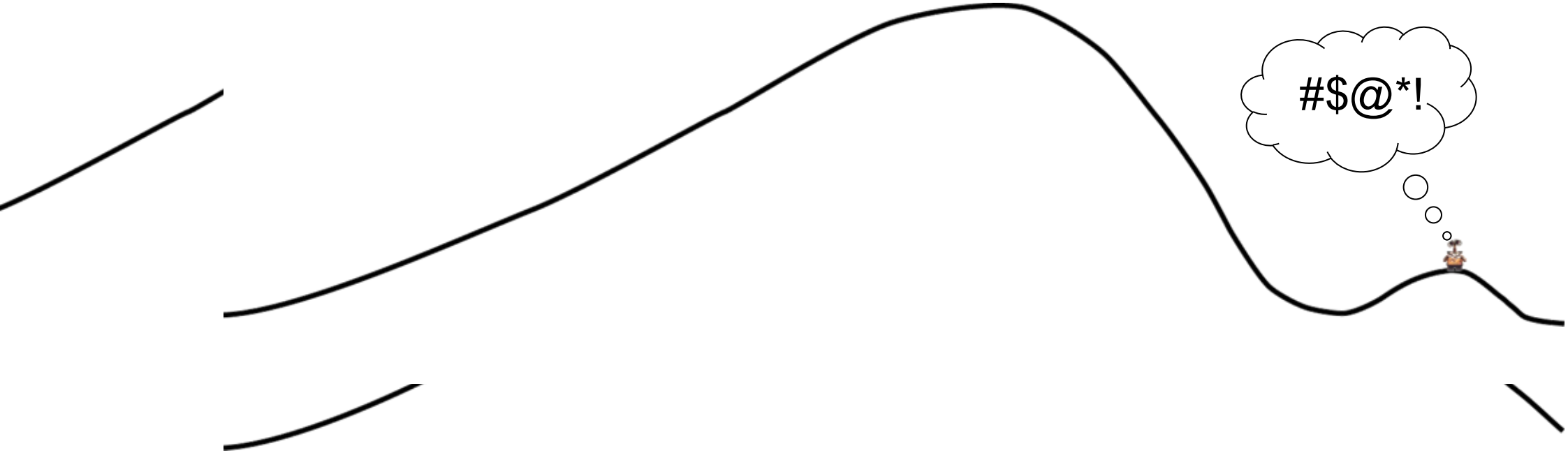
Maximum likelihood phylogeny

- **Imagine tree 'space' is a hill**
- **Better trees (measured by likelihood) are higher**
- **We can find the best tree using a robot with a simple program:**
  - Accept uphill moves
  - Reject downhill moves



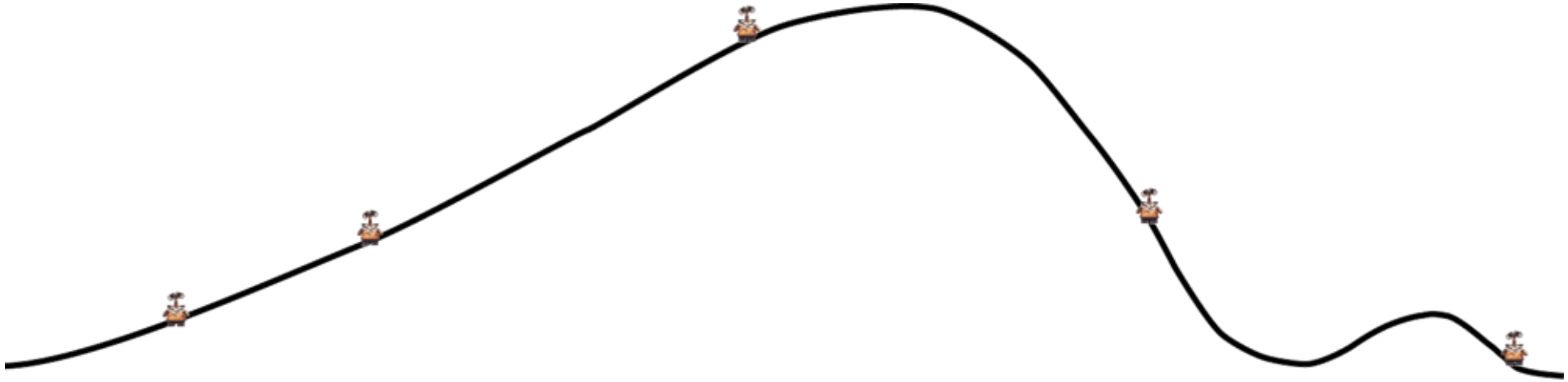
Courtesy of Sanger Genomic course

Hill Climbing



Hill Climbing

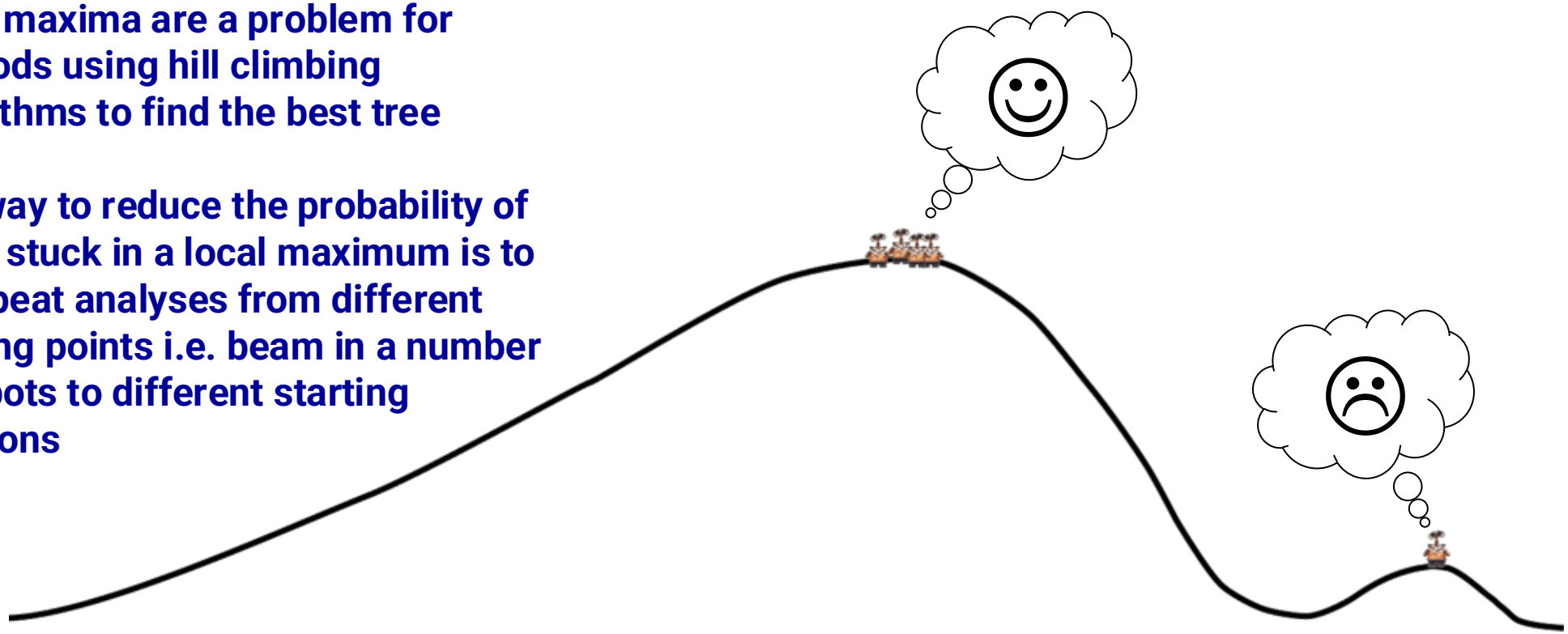
- Local maxima are a problem for methods using hill climbing algorithms to find the best tree
- One way to reduce the probability of being stuck in a local maximum is to do repeat analyses from different starting points
- I.e. beam in a number of robots to different starting positions



Hill Climbing

- Local maxima are a problem for methods using hill climbing algorithms to find the best tree

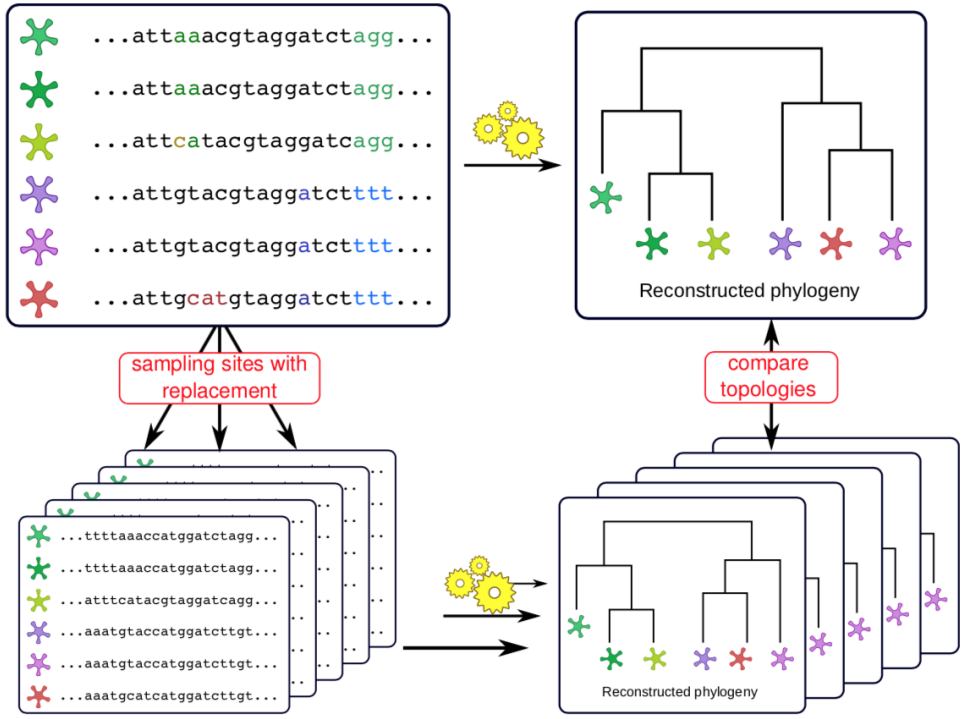
- One way to reduce the probability of being stuck in a local maximum is to do repeat analyses from different starting points i.e. beam in a number of robots to different starting positions



Bootstrapping

How much do you trust the tree that you just built?

Assess variability due to sampling and conflicting signals, relying on analyzing resampled dataset.

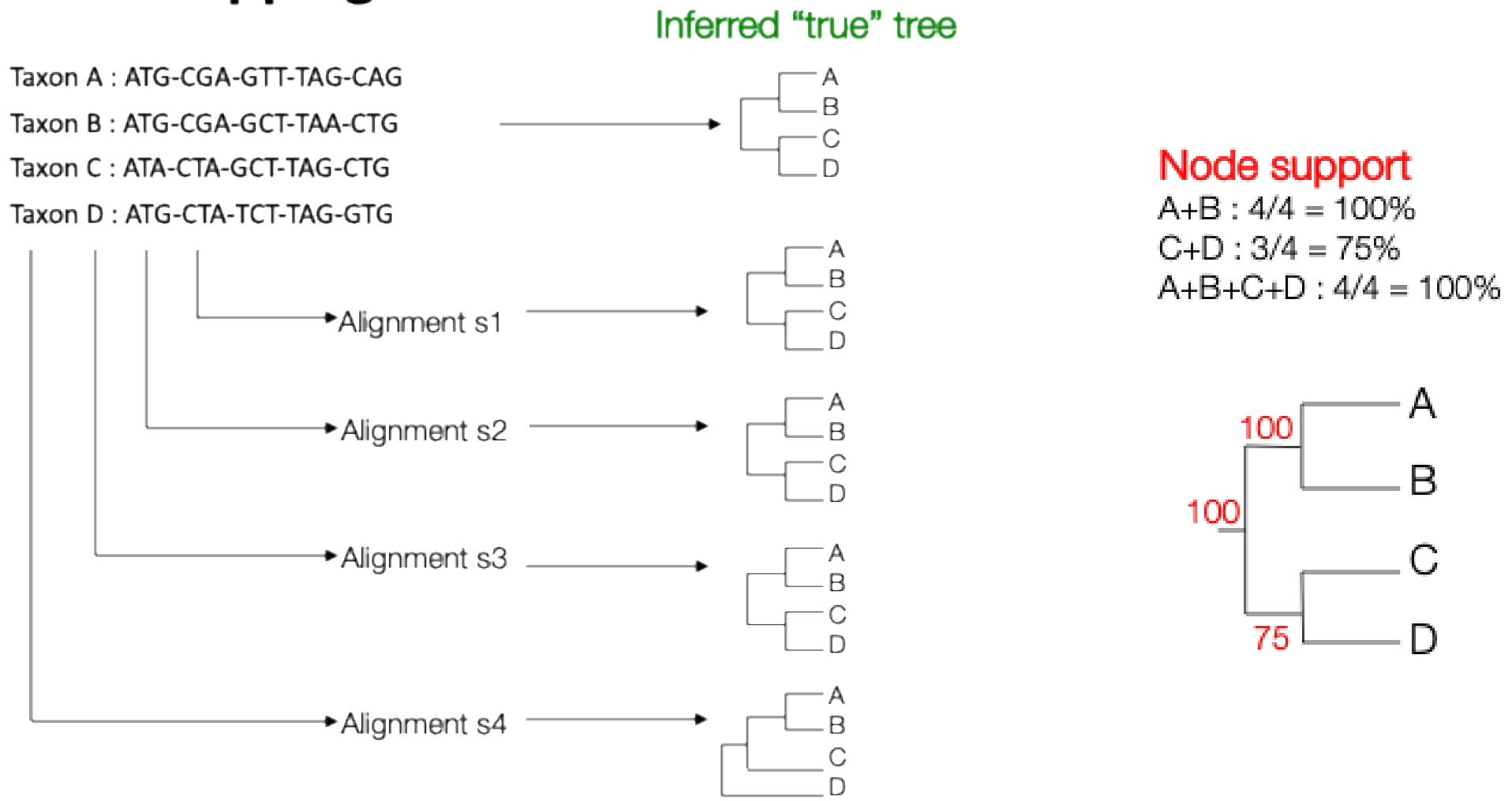


Taxa	Characters 1	2	3	4	5	6	7	8	9
A	A	C	C	T	G	A	T	G	C
B	A	G	C	T	G	G	T	T	C
C	A	G	C	A	G	A	T	G	G
D	T	C	C	T	C	G	T	G	C
E	T	C	T	T	A	A	T	G	C

Permutation with replacement

Taxa	Characters 2	5	9	2	7	7	2	1	6
A	C	G	C	C	T	T	C	A	A
B	G	G	C	G	T	T	G	A	G
C	G	G	G	G	T	T	G	A	A
D	C	C	C	C	T	T	C	T	G
E	C	A	C	C	T	T	C	T	A

Bootstrapping



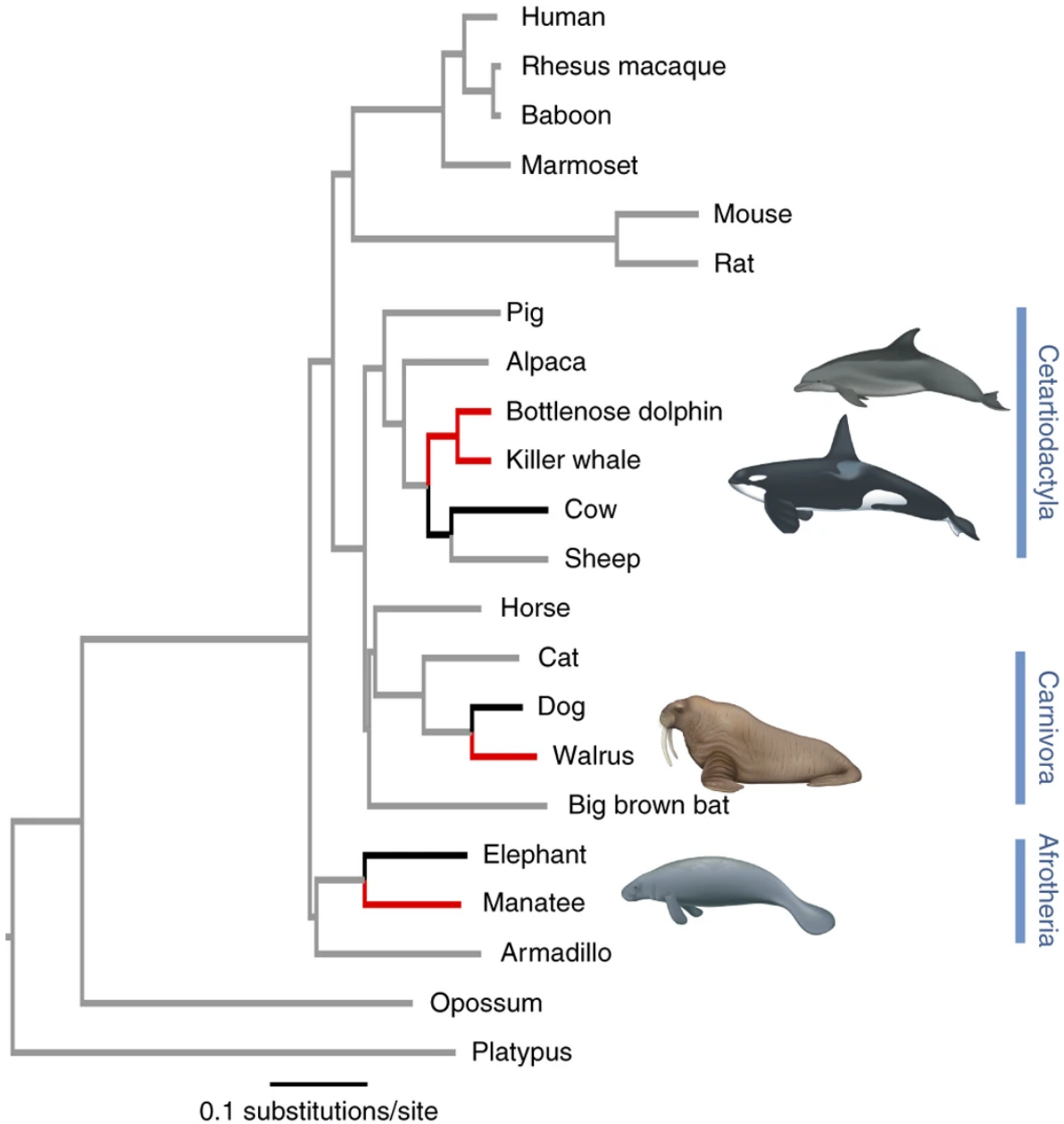
A tree without measures of statistical support for the nodes (bootstraps or posteriors) is uncertain!

Summary

- Phylogenetic trees are (nearly) best model to represent genetic relatedness and evolutionary history of a groups of organisms
- Trees are built using **homologous** sequence alignments -> most important input
- Unrooted trees provide information about relatedness, while rooted trees tell more complete story of evolution.
- Most publications use maximum likelihood as a default phylogenetic construction tool, which in most cases provide suitable inferences
- Tree building is automated by softwares, so **DNA alignment is most important**.
- A good tree is a tree with bootstraps and a clear root.

Tree reading exercise




Human
Rhesus macaque
Baboon
Marmoset
Mouse
Rat
Pig
Alpaca
Bottlenose dolphin
Killer whale
Cow
Sheep
Horse
Cat
Dog
Walrus
Big brown bat
Elephant
Manatee
Armadillo
Opossum
Platypus
0.1 substitutions/site
Cetartiodactyla
Carnivora
Afrotheria

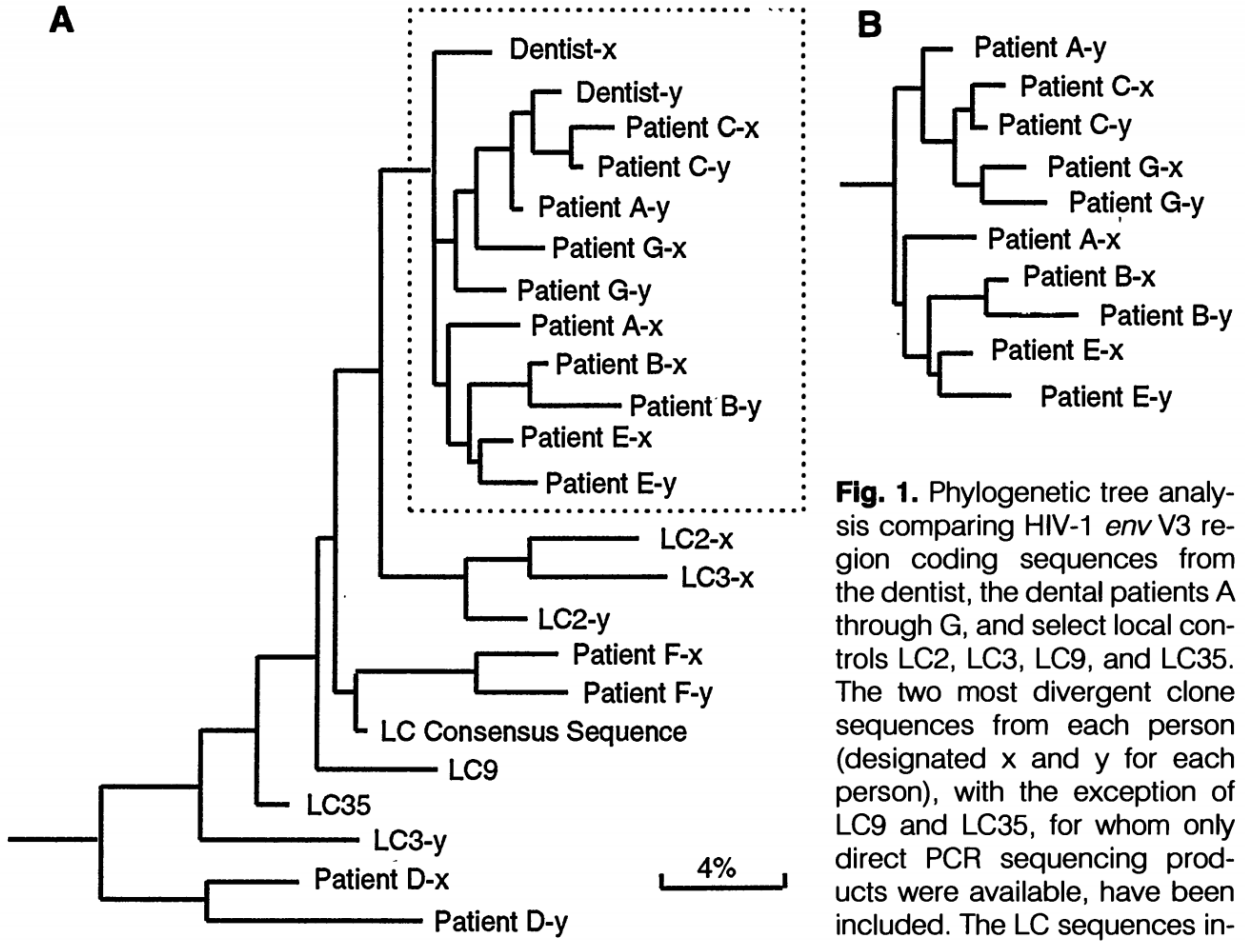
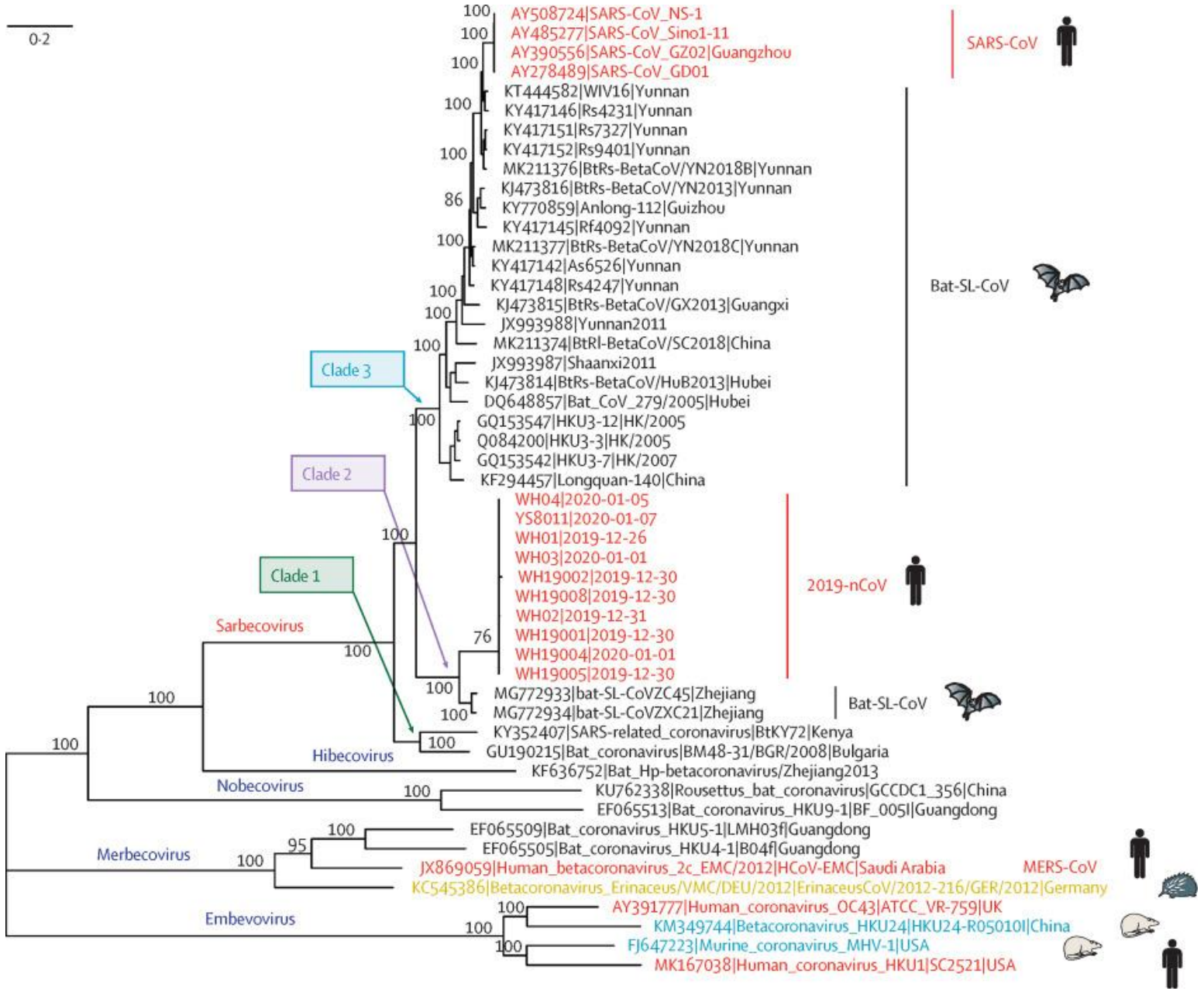
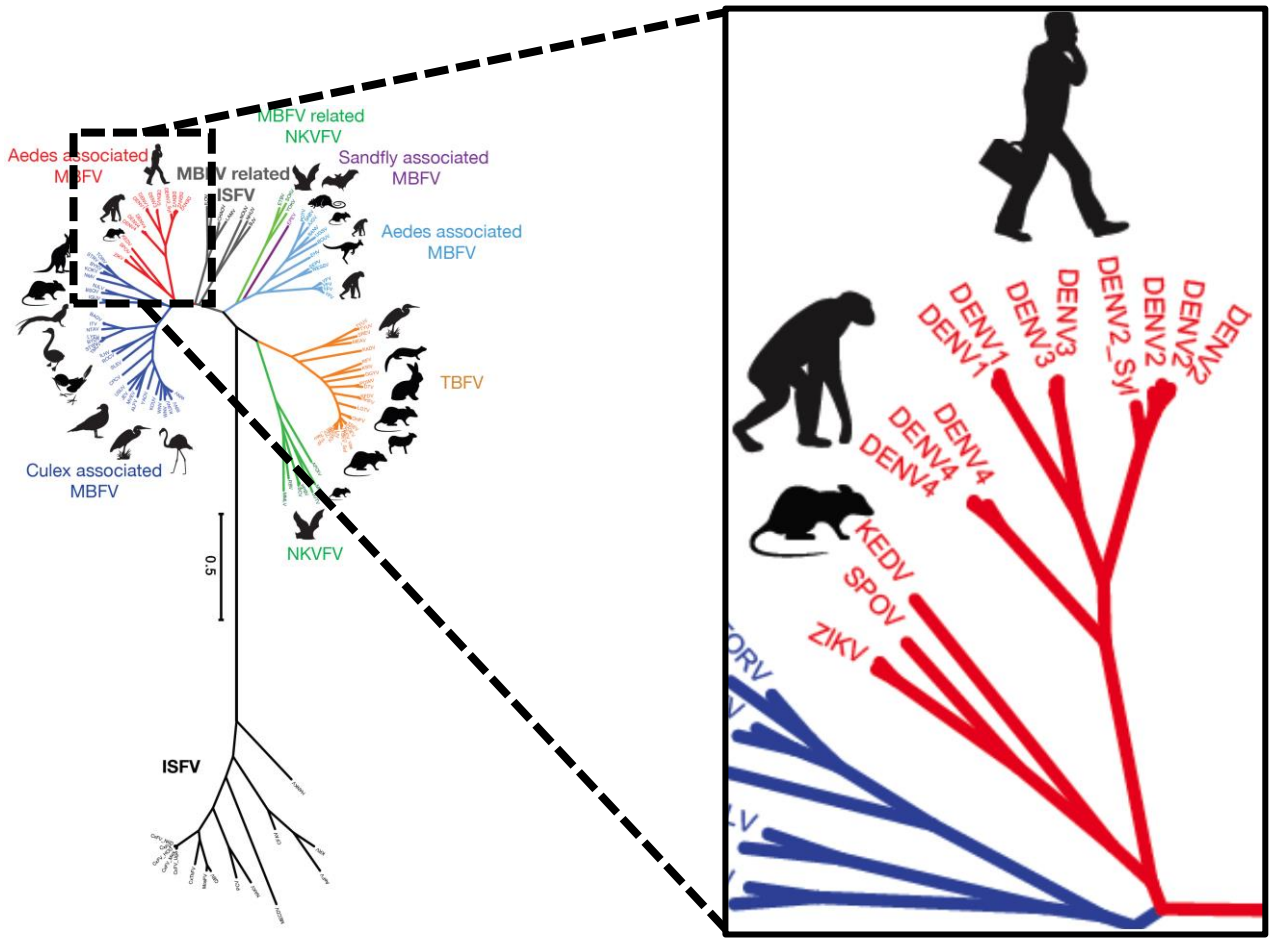
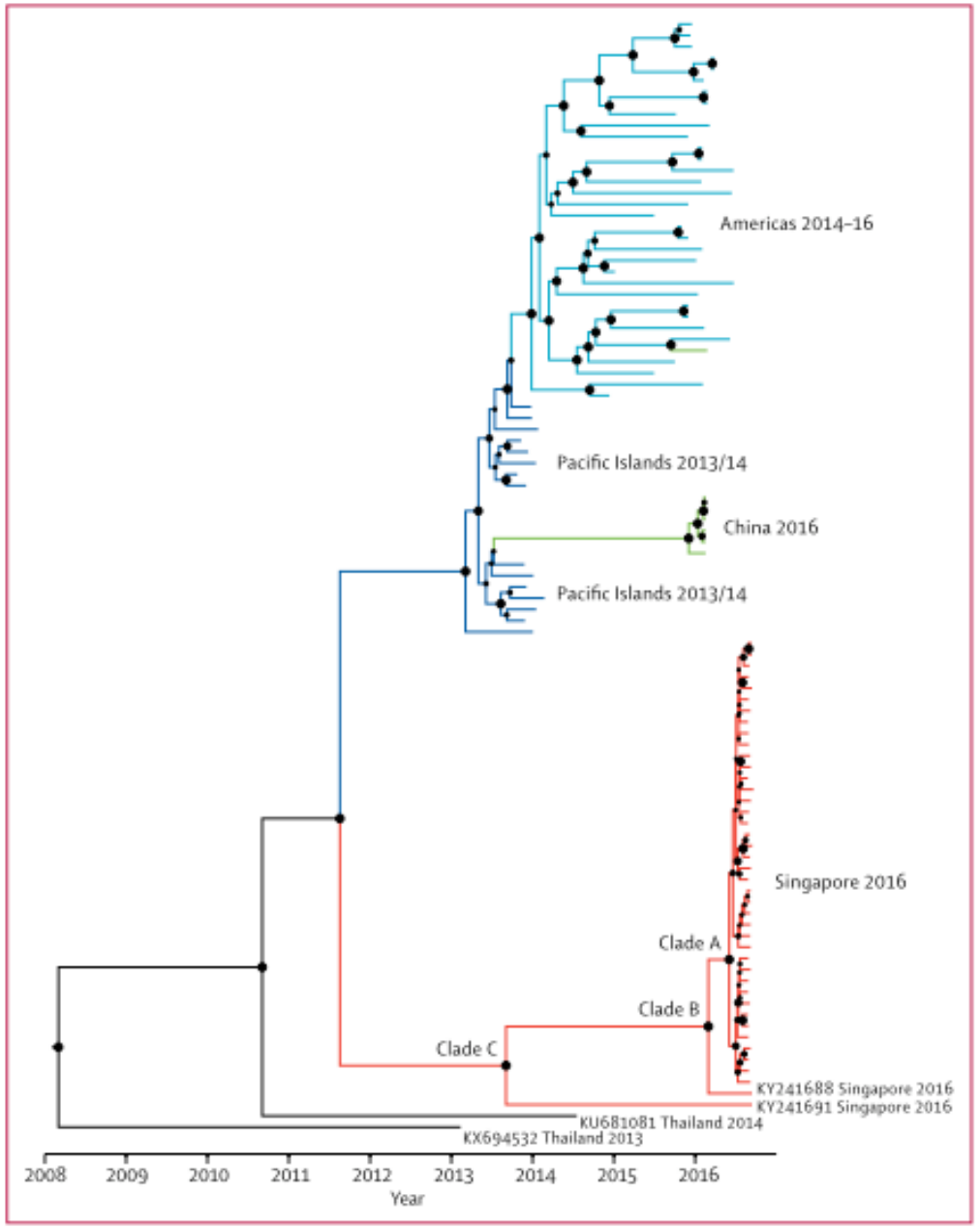


Fig. 1. Phylogenetic tree analysis comparing HIV-1 *env* V3 region coding sequences from the dentist, the dental patients A through G, and select local controls LC2, LC3, LC9, and LC35. The two most divergent clone sequences from each person (designated x and y for each person), with the exception of LC9 and LC35, for whom only direct PCR sequencing products were available, have been included. The LC sequences included were those found by pairwise distance measurement (Table 1) and signature pattern analysis (Table 2) to be the closest control sequences to the dental group sequences, which are enclosed by a box in (**A**). An LC consensus sequence has also been included, and the tree was rooted upon the African sample ELI. The PAUP parsimony algorithm was used to analyze 279 aligned sites (*25*), of which 146 sites were varied. When the dentist's viral sequences were withdrawn from the analysis, or required to cluster with the LC consensus sequence (*25*), the dental clade remained otherwise unaffected, as shown in (**B**). Vertical distances are for clarity only; the lengths of the horizontal branches are proportional to the single base changes and can be read as percentage differences with the scale bar.


0·2
AY508724|SARS-CoV_NS-1
AY485277|SARS-CoV_Sino1-11
AY390556|SARS-CoV_GZ02|Guangzhou
AY278489|SARS-CoV_GD01
SARS-CoV
KT444582|WIV16|Yunnan
KY417146|Rs4231|Yunnan
KY417151|Rs7327|Yunnan
KY417152|Rs9401|Yunnan
MK211376|BtRs-BetaCoV/YN2018B|Yunnan
KJ473816|BtRs-BetaCoV/YN2013|Yunnan
KY770859|Anlong-112|Guizhou
KY417145|Rf4092|Yunnan
MK211377|BtRs-BetaCoV/YN2018C|Yunnan
KY417142|As6526|Yunnan
KY417148|Rs4247|Yunnan
KJ473815|BtRs-BetaCoV/GX2013|Guangxi
JX993988|Yunnan2011
MK211374|BtRl-BetaCoV/SC2018|China
JX993987|Shaanxi2011
KJ473814|BtRs-BetaCoV/HuB2013|Hubei
DQ648857|Bat_CoV_279/2005|Hubei
GQ153547|HKU3-12|HK/2005
Q084200|HKU3-3|HK/2005
GQ153542|HKU3-7|HK/2007
KF294457|Longquan-140|China
Bat-SL-CoV
Clade 3
Clade 2
Clade 1
WH04|2020-01-05
YS8011|2020-01-07
WH01|2019-12-26
WH03|2020-01-01
WH19002|2019-12-30
WH19008|2019-12-30
WH02|2019-12-31
WH19001|2019-12-30
WH19004|2020-01-01
WH19005|2019-12-30
2019-nCoV
Sarbecovirus
MG772933|bat-SL-CoVZC45|Zhejiang
MG772934|bat-SL-CoVZXC21|Zhejiang
Bat-SL-CoV
KY352407|SARS-related_coronavirus|BtKY72|Kenya
GU190215|Bat_coronavirus|BM48-31/BGR/2008|Bulgaria
Hibecovirus
KF636752|Bat_Hp-betacoronavirus/Zhejiang2013
Nobecovirus
KU762338|Rousettus_bat_coronavirus|GCCDC1_356|China
EF065513|Bat_coronavirus_HKU9-1|BF_005I|Guangdong
EF065509|Bat_coronavirus_HKU5-1|LMH03f|Guangdong
Merbecovirus
EF065505|Bat_coronavirus_HKU4-1|B04f|Guangdong
JX869059|Human_betacoronavirus_2c_EMC/2012|HCoV-EMC|Saudi Arabia
MERS-CoV
KC545386|Betacoronavirus_Erinaceus/VMC/DEU/2012|ErinaceusCoV/2012-216/GER/2012|Germany
AY391777|Human_coronavirus_OC43|ATCC_VR-759|UK
Embevovirus
KM349744|Betacoronavirus_HKU24|HKU24-R05010I|China
FJ647223|Murine_coronavirus_MHV-1|USA
MK167038|Human_coronavirus_HKU1|SC2521|USA



Flaviviridae family of viruses



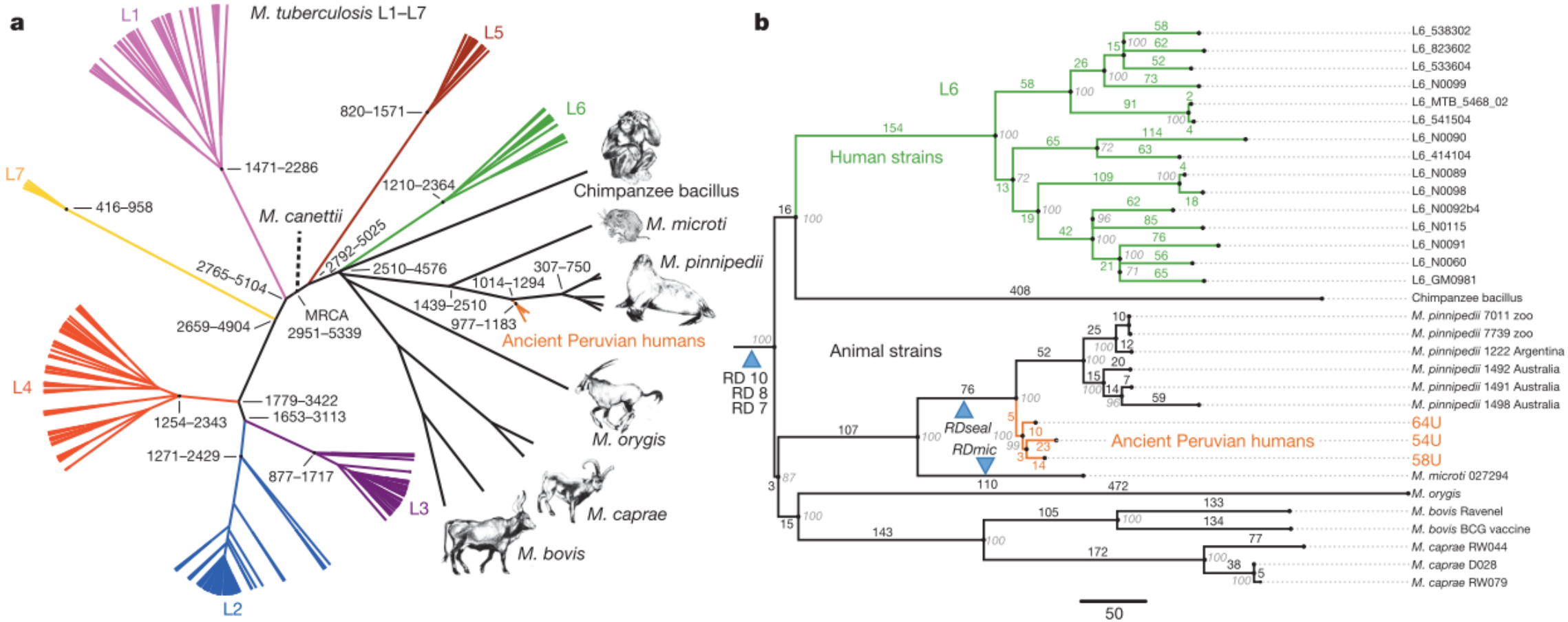
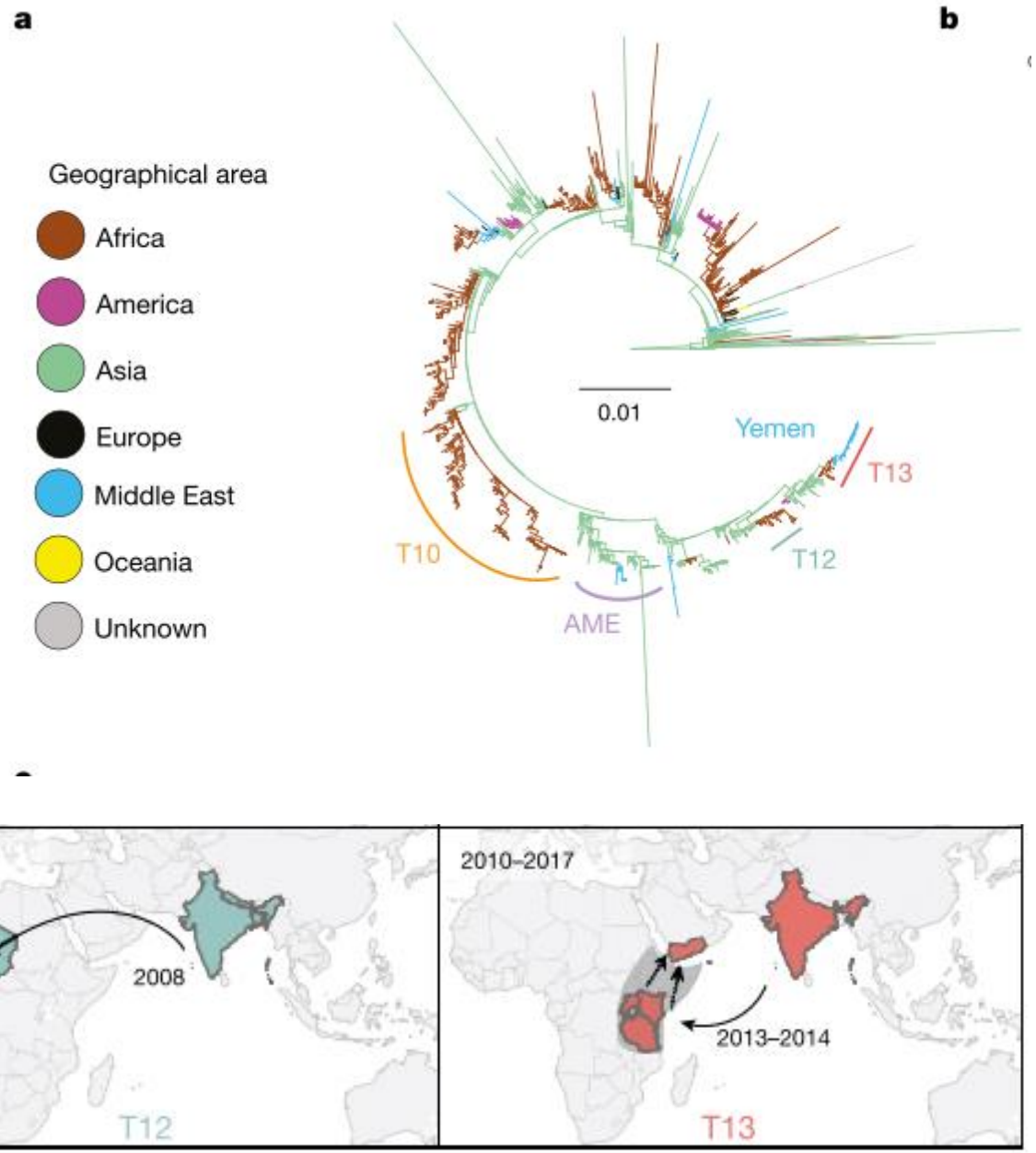
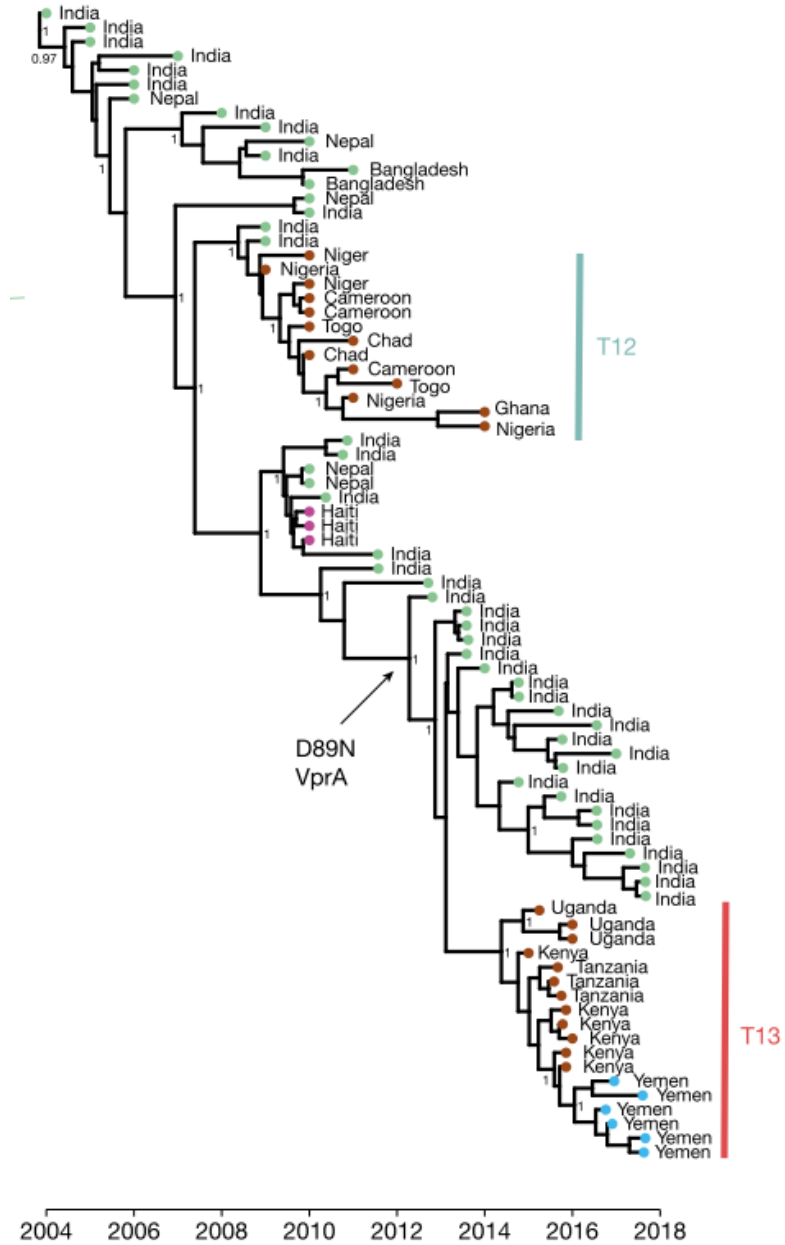
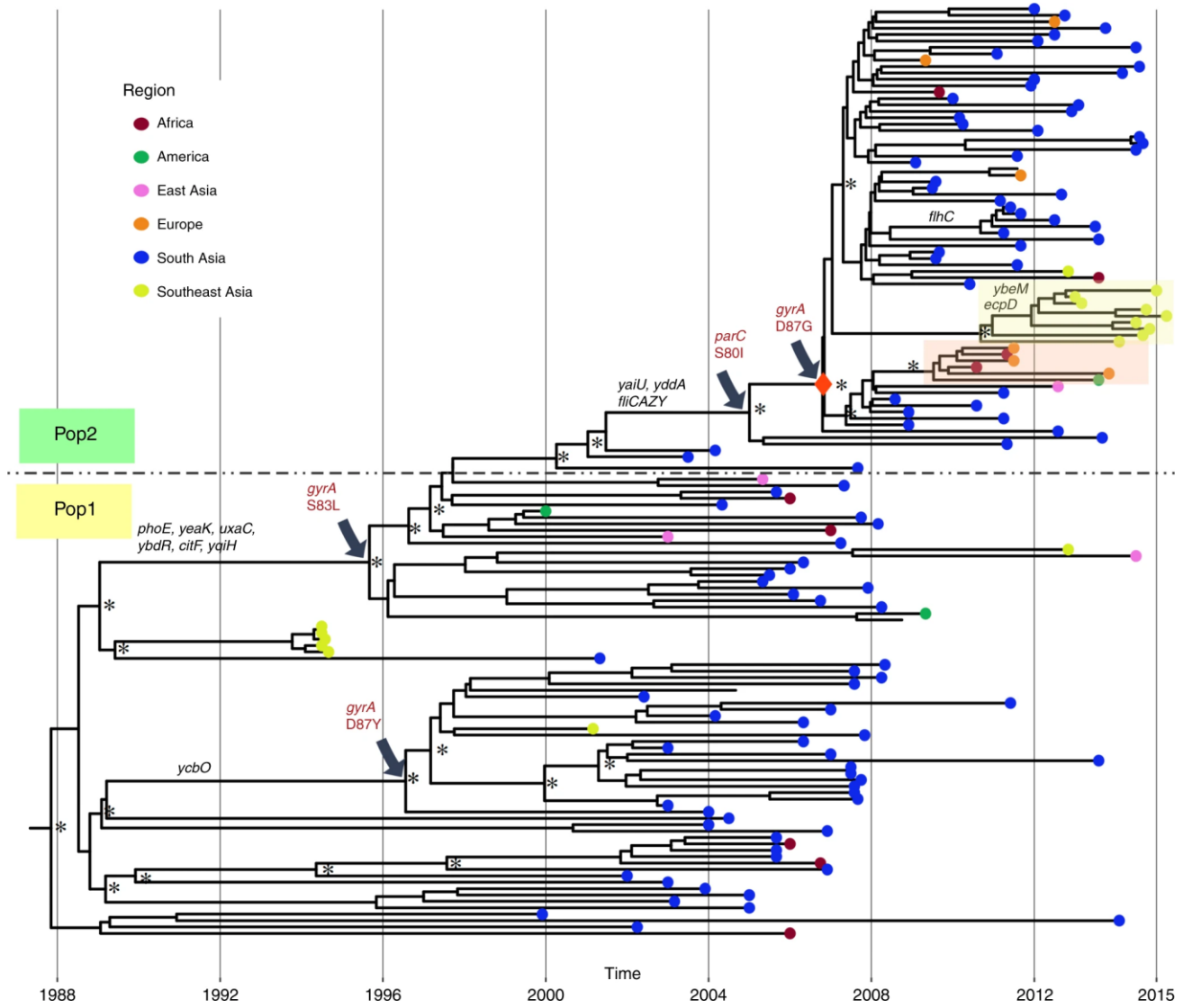


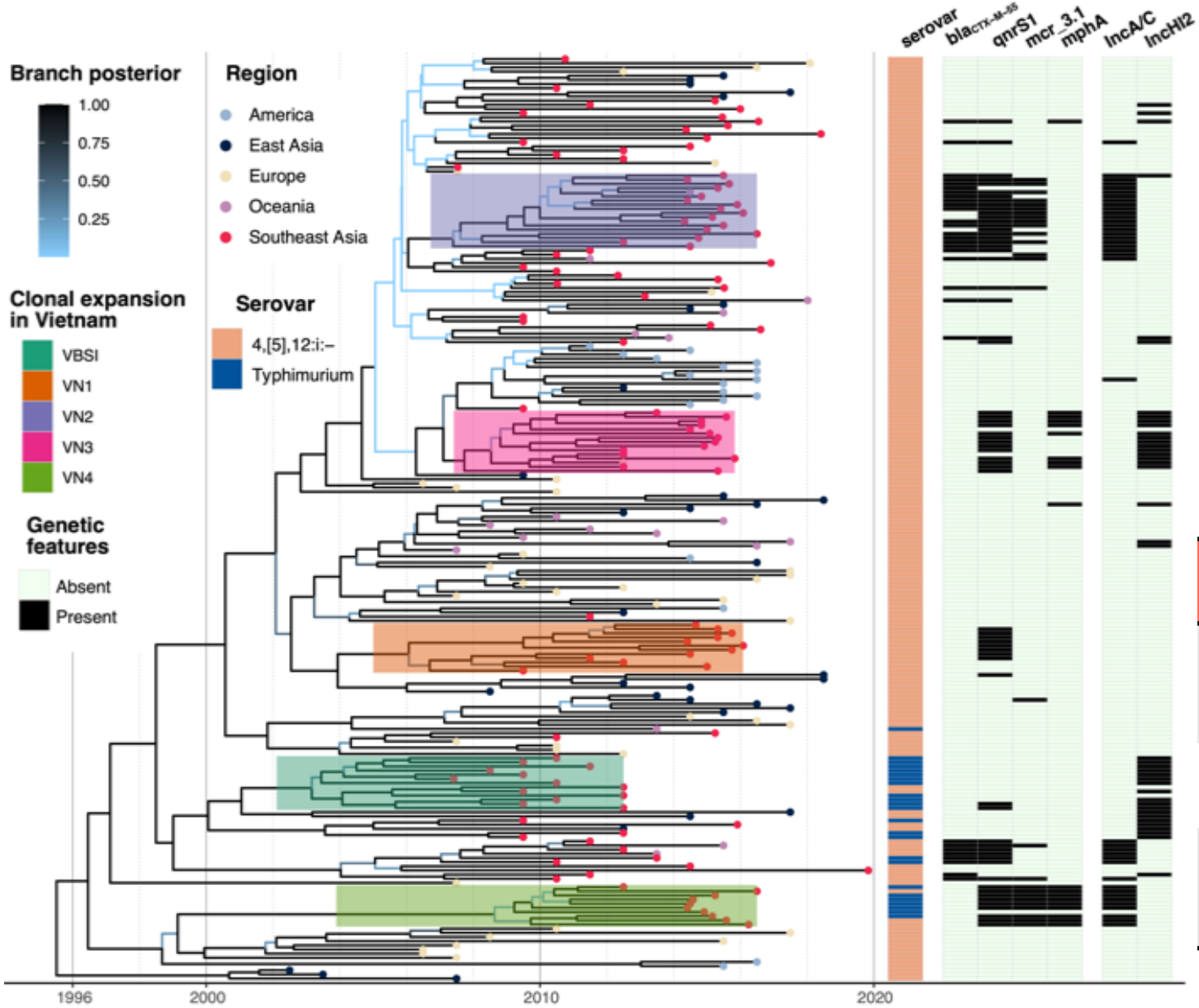
Figure 3 | Phylogenetic analysis. **a**, Bayesian maximum clade credibility tree of 261 MTBC genomes (excluding Hungarian mummy), with estimated divergence dates shown in years before present using a model of population expansion. **b**, Maximum parsimony tree for lineage 6 and animal-adapted MTBC genomes with SNPs that define all branches. Bootstrap values in grey italics. Deletions specific to the animal lineages are shown as triangles.


a
Geographical area
Africa
America
Asia
Europe
Middle East
Oceania
Unknown
b
0.01
Yemen
T13
T12
T10
AME
2010–2017
2008
T12
2010–2017
2013–2014
T13


India
Nepal
Bangladesh
Niger
Nigeria
Cameroon
Togo
Chad
Ghana
Haiti
Uganda
Kenya
Tanzania
Yemen
T12
T13
D89N
VprA
2004
2006
2008
2010
2012
2014
2016
2018



Temporal phylogenetic reconstruction of CenAsiaIII *Shigella sonnei* overlaid with pseudogenes. A maximum clade credibility phylogenetic reconstruction of 144 CenAsiaIII *S. sonnei* is shown. Asterisks indicate posterior probability support >80% on major internal nodes. The red diamond indicates the internal node leading to the major fluoroquinolone-resistant (FQr) clone. Grey arrows designate branches on which certain quinolone resistance determining region (QRDR) mutations occurred.



clone	VBSI	VN1	VN2	VN3	VN4
tMRCA	2003				
AMR					
plasmid					



Thank you!

Question?

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